

VITYARTHI PROJECT- REPORT

“Donor Hub”

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INTRODUCTION:

In an era where healthcare advancements continue to push boundaries, the fundamental act of blood donation remains one of the most critical and life-sustaining procedures in modern medicine. Every two seconds, someone in the world needs blood, yet the gap between supply and demand continues to widen. **THE DONOR HUB** emerges as a transformative solution at the intersection of healthcare technology and humanitarian service, designed to revolutionize how we approach blood donor eligibility and screening.

In developing **THE DONOR HUB**, we recognize that blood donation is a cornerstone of emergency healthcare, surgical procedures, cancer treatments, and chronic disease management. Our platform serves as a critical enabler in the healthcare value chain, supporting not just individual donors and blood banks, but ultimately the patients whose lives depend on timely access to safe blood products.

PROBLEM STATEMENT:

Blood donation is a critical healthcare service that saves millions of lives annually. However, the process of determining donor eligibility is often manual, time-consuming, and prone to human error. Many potential donors are unaware of the comprehensive medical criteria required for safe blood donation, leading to:

- **Inefficient screening processes** at blood donation camps
- **Lack of awareness** among potential donors about eligibility requirements
- **Inconsistent application** of medical criteria across different donation centers
- **Time wastage** for both donors and healthcare staff when ineligible donors arrive
- **Potential health risks** to both donors and recipients when proper screening isn't followed

This creates a significant gap in the blood donation ecosystem where willing donors may be turned away after investing time and effort, while blood banks struggle with inefficient screening processes.

Functional Requirements:

1. User Registration & Authentication Requirements
 - Donor registration
 - Input Validation and error handling
2. Medical Screening & Eligibility Assessment Requirements
 - Vital parameters assessment
 - Medical history screening
 - Gender-specific screening
3. Eligibility Engine Requirements
 - securitRule based eligibility determination

Non-Functional Requirements:

1. Performance Requirements
 - Response time requirements
 - Resource utilization
2. reliability and availability requirements
 - System reliability
 - Data integrity
3. Security requirements
 - Data security and privacy
 - Compliance
4. Usability requirements

- User experience
 - Accessibility
5. Maintability requirements
- Code quality
 - Supportability
6. Scalability requirements
- Growth capacity
 - Performance under load

Source code:

```
import re
```

```
class DonorHub:
```

```
    def __init__(self):
```

```
        self.diseases = [
```

```
            "HIV Infection",
```

```
            "Hepatitis B or C",
```

```
            "Convulsions and Epilepsy",
```

```
            "Cardio Vascular Diseases",
```

```
            "Schizophrenia",
```

```
            "Chronic Liver Disease",
```

```
            "Current Fever / Acute Infection",
```

```
            "Skin Diseases at donation site",
```

```
            "Autoimmune Disorders",
```

```
            "Severe Allergic Disorders",
```

```
            "Recipients of Transplants",
```

```
            "Gonorrhoea, Leprosy, Leishmaniasis, Conjunctivitis (recent)"
```

]

def display_title(self):

print("=" * 60)

print(" THE DONOR HUB")

print("=" * 60)

print("Welcome to donor hub. Register now and save lives!")

print()

def get_personal_info(self):

print("PERSONAL INFORMATION")

print("-" * 30)

name = input("Enter your full name: ")

while True:

email = input("Enter your email: ")

if self.validate_email(email):

break

else:

print("Invalid email format. Please try again.")

while True:

phone = input("Enter your phone number: ")

if self.validate_phone(phone):

break

else:

print("Invalid phone number. Please enter 10 digits.")

```
address = input("Enter your address: ")
```

```
while True:
```

```
    gender = input("Enter your gender (Male/Female): ").strip().lower()
```

```
    if gender in ['male', 'female']:
```

```
        break
```

```
    else:
```

```
        print("Please enter either 'Male' or 'Female'")
```

```
while True:
```

```
    first_time = input("Is this your first time donating? (Yes/No): ").strip().lower()
```

```
    if first_time in ['yes', 'no']:
```

```
        break
```

```
    else:
```

```
        print("Please enter either 'Yes' or 'No'")
```

```
return {
```

```
    'name': name,
```

```
    'email': email,
```

```
    'phone': phone,
```

```
    'address': address,
```

```
    'gender': gender,
```

```
    'first_time': first_time
```

```
}
```

```
def validate_email(self, email):
```

```
    pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
```

```
    return re.match(pattern, email) is not None
```

```
def validate_phone(self, phone):
```

```
    return phone.isdigit() and len(phone) == 10
```

```
def get_medical_info(self):
```

```
    print("\nMEDICAL INFORMATION")
```

```
    print("-" * 30)
```

```
while True:
```

```
    try:
```

```
        age = int(input("Enter your age: "))
```

```
        if 18 <= age <= 100:
```

```
            break
```

```
        else:
```

```
            print("Age must be between 18 and 100. Please try again.")
```

```
    except ValueError:
```

```
        print("Please enter a valid number for age.")
```

```
while True:
```

```
    try:
```

```
        weight = float(input("Enter your weight (kg): "))
```

```
        if 30 <= weight <= 200:
```

```
            break
```

```
        else:
```

```
            print("Weight must be between 30 and 200 kg. Please try again.")
```

```
    except ValueError:
```

```
        print("Please enter a valid number for weight.")
```

```
while True:
```

```
try:

    hemoglobin = float(input("Enter your hemoglobin level (g/dL): "))

    if 5 <= hemoglobin <= 20:

        break

    else:

        print("Hemoglobin must be between 5 and 20 g/dL. Please try again.")

except ValueError:

    print("Please enter a valid number for hemoglobin.")


blood_group = input("Enter your blood group (e.g., A+, O-, AB+, etc.): ")
blood_group.upper()


while True:

    blood_pressure = input("Enter your blood pressure (e.g., 120/80): ")

    if self.validate_blood_pressure(blood_pressure):

        break

    else:

        print("Invalid blood pressure format. Please use format like '120/80'")


return {

    'age': age,

    'weight': weight,

    'hemoglobin': hemoglobin,

    'blood_group': blood_group,

    'blood_pressure': blood_pressure

}


def validate_blood_pressure(self, bp):

    pattern = r'^\d{2,3}/\d{2,3}$'
```

```

if re.match(pattern, bp):

    systolic, diastolic = map(int, bp.split('/'))

    return 60 <= systolic <= 200 and 40 <= diastolic <= 150

return False


def get_disease_history(self):

    print("\nMEDICAL HISTORY - Please tick (enter 'yes') for any conditions you
have:")

    print("-" * 60)


    selected_diseases = []

    for disease in self.diseases:

        while True:

            response = input(f"Do you have {disease}? (Yes/No): ").strip().lower()

            if response in ['yes', 'no']:

                if response == 'yes':

                    selected_diseases.append(disease)

                    break

            else:

                print("Please enter either 'Yes' or 'No'")


    return selected_diseases


def get_female_specific_info(self):

    print("\nADDITIONAL INFORMATION FOR FEMALE DONORS")

    print("-" * 40)


    while True:

        pregnant = input("Are you currently pregnant? (Yes/No): ").strip().lower()

```

```

        if pregnant in ['yes', 'no']:
            break
        else:
            print("Please enter either 'Yes' or 'No'")

    while True:
        abortion = input("Have you had an abortion in the last 6 months? (Yes/No):
").strip().lower()
        if abortion in ['yes', 'no']:
            break
        else:
            print("Please enter either 'Yes' or 'No'")

    while True:
        menstruating = input("Are you currently menstruating? (Yes/No):
").strip().lower()
        if menstruating in ['yes', 'no']:
            break
        else:
            print("Please enter either 'Yes' or 'No'")

    return {
        'pregnant': pregnant,
        'abortion': abortion,
        'menstruating': menstruating
    }

def check_eligibility(self, personal_info, medical_info, diseases,
female_info=None):
    print("\n" + "=" * 60)

```



```

# Display criteria check

print(f"✓ Age check (18-65): {'PASS' if age_eligible else 'FAIL'}")

print(f"✓ Weight check (45-80 kg): {'PASS' if weight_eligible else 'FAIL'}")

print(f"✓ Hemoglobin check (≥12.5 g/dL): {'PASS' if hemoglobin_eligible else 'FAIL'}")

print(f"✓ Blood pressure check (approx 120/80): {'PASS' if bp_eligible else 'FAIL'}")

print(f"✓ Medical history check (no diseases): {'PASS' if disease_eligible else 'FAIL'}")

if personal_info['gender'] == 'female':

    print(f"✓ Female-specific criteria: {'PASS' if female_eligible else 'FAIL'}")

# Final eligibility

if (age_eligible and weight_eligible and hemoglobin_eligible and
    bp_eligible and disease_eligible and female_eligible):

    print("\n🎉 CONGRATULATIONS! You are ELIGIBLE for blood donation! 🎉")

    return True

else:

    print("\n❌ Unfortunately, you are NOT ELIGIBLE for blood donation at this time.")

    return False

def display_precautions(self):

    print("\n" + "=" * 60)

    print("IMPORTANT PRECAUTIONS FOR BLOOD DONATION")

    print("=" * 60)

    precautions = [

        "1. The donor shall be in good health, mentally alert and physically fit and
        shall not be inmates of jail or any other confinement.",

```

"2. First time donor shall not be over 60 years of age, for repeat donor upper limit is 65 years.",

"3. The donor shall not be fasting before the blood donation or observing fast during the period of blood donation and last meal should have been taken at least 4 hours prior to donation.",

"4. Donor shall not have consumed alcohol and show signs of intoxication before the blood donation. The donor shall not be a person having regular heavy alcohol intake.",

"5. The donor who works as air crew member, long distance vehicle driver, either above sea level or below sea level or in emergency services or where strenuous work is required, shall not donate blood at least 24 hours prior to their next duty shift. The donor shall not be a night shift workers without adequate sleep."

]

for precaution in precautions:

print(precaution)

def run(self):

self.display_title()

Get personal information

personal_info = self.get_personal_info()

Get medical information

medical_info = self.get_medical_info()

Get disease history

diseases = self.get_disease_history()

Get female-specific information if applicable

female_info = None

```
if personal_info['gender'] == 'female':  
    female_info = self.get_female_specific_info()  
  
# Check eligibility  
is_eligible = self.check_eligibility(personal_info, medical_info, diseases,  
female_info)  
  
# Display precautions  
self.display_precautions()  
  
print("\n" + "=" * 60)  
print("Thank you for your interest in saving lives!")  
print("=" * 60)  
  
# Run the program  
if __name__ == "__main__":  
    donor_hub = DonorHub()  
    donor_hub.run()
```

Output Screenshots:

```
C:\Windows\system32\cmd.exe: X + v

C:\Users\palak\important shi-\python\New folder>python project.py
=====
THE DONOR HUB
=====
Welcome to donor hub. Register now and save lives!

PERSONAL INFORMATION
=====
Enter your full name: palak roy
Enter your email: palak@gmail.com
Enter your phone number: 4587859455
Enter your address: vit bhopal
Enter your gender (Male/Female): female
Is this your first time donating? (Yes/No): yes

MEDICAL INFORMATION
=====
Enter your age: 19
Enter your weight (kg): 65
Enter your hemoglobin level (g/dL): 15
Enter your blood group (e.g., A+, O-, AB+, etc.): O+
Enter your blood pressure (e.g., 120/80): 120/80

MEDICAL HISTORY - Please tick (enter 'yes') for any conditions you have:
=====
Do you have HIV Infection? (Yes/No): no
Do you have Hepatitis B or C? (Yes/No): no
Do you have Convulsions and Epilepsy? (Yes/No): no
Do you have Cardio Vascular Diseases? (Yes/No): no
Do you have Schizophrenia? (Yes/No): no
Do you have Chronic Liver Disease? (Yes/No): no
Do you have Current Fever / Acute Infection? (Yes/No): no
Do you have Skin Diseases at donation site? (Yes/No): no
Do you have Autoimmune Disorders? (Yes/No): no
Do you have Severe Allergic Disorders? (Yes/No): no
Do you have Recipients of Transplants? (Yes/No): no
Do you have Gonorrhoea, Leprosy, Leishmaniasis, Conjunctivitis (recent)? (Yes/No): no

ADDITIONAL INFORMATION FOR FEMALE DONORS
=====
Are you currently pregnant? (Yes/No): no
Have you had an abortion in the last 6 months? (Yes/No): no
Are you currently menstruating? (Yes/No): yes

Do you have Schizophrenia? (Yes/No): no
Do you have Chronic Liver Disease? (Yes/No): no
Do you have Current Fever / Acute Infection? (Yes/No): no
Do you have Skin Diseases at donation site? (Yes/No): no
Do you have Autoimmune Disorders? (Yes/No): no
Do you have Severe Allergic Disorders? (Yes/No): no
Do you have Recipients of Transplants? (Yes/No): no
Do you have Gonorrhoea, Leprosy, Leishmaniasis, Conjunctivitis (recent)? (Yes/No): no

ADDITIONAL INFORMATION FOR FEMALE DONORS
=====
Are you currently pregnant? (Yes/No): no
Have you had an abortion in the last 6 months? (Yes/No): no
Are you currently menstruating? (Yes/No): yes

=====
ELIGIBILITY CHECK RESULTS
=====
✓ Age check (18-65): PASS
✓ Weight check (45-80 kg): PASS
✓ Hemoglobin check ( $\geq 12.5$  g/dL): PASS
✓ Blood pressure check (approx 120/80): PASS
✓ Medical history check (no diseases): PASS
✓ Female-specific criteria: FAIL

✗ Unfortunately, you are NOT ELIGIBLE for blood donation at this time.

=====
IMPORTANT PRECAUTIONS FOR BLOOD DONATION
=====
1. The donor shall be in good health, mentally alert and physically fit and shall not be inmates of jail or any other confinement.
2. First time donor shall not be over 60 years of age, for repeat donor upper limit is 65 years.
3. The donor shall not be fasting before the blood donation or observing fast during the period of blood donation and last meal should have been taken at least 4 hours prior to donation.
4. Donor shall not have consumed alcohol and show signs of intoxication before the blood donation. The donor shall not be a person having regular heavy alcohol intake.
5. The donor who works as air crew member, long distance vehicle driver, either above sea level or below sea level or in emergency services or where strenuous work is required, shall not donate blood at least 24 hours prior to their next duty shift. The donor shall not be a night shift workers without adequate sleep.

=====
Thank you for your interest in saving lives!
=====
```

```
C:\Windows\system32\cmd.exe
C:\Users\palak\important shi-\python\New folder>python project.py
=====
THE DONOR HUB
=====
Welcome to donor hub. Register now and save lives!

PERSONAL INFORMATION
=====
Enter your full name: palak
Enter your email: palak@gmail.com
Enter your phone number: 4587825455
Enter your address: vit bhopal
Enter your gender (Male/Female): female
Is this your first time donating? (Yes/No): no

MEDICAL INFORMATION
=====
Enter your age: 45
Enter your weight (kg): 70
Enter your hemoglobin level (g/dL): 14
Enter your blood group (e.g., A+, O-, AB+, etc.): A+
Enter your blood pressure (e.g., 120/80): 120/80

MEDICAL HISTORY - Please tick (enter 'yes') for any conditions you have:
=====
Do you have HIV Infection? (Yes/No): no
Please enter either 'Yes' or 'No'
Do you have HIV Infection? (Yes/No): no
Do you have Hepatitis B or C? (Yes/No): no
Do you have Convulsions and Epilepsy? (Yes/No): no
Do you have Cardio Vascular Diseases? (Yes/No): no
Do you have Schizophrenia? (Yes/No): no
Do you have Chronic Liver Disease? (Yes/No): no
Do you have Current Fever / Acute Infection? (Yes/No): no
Do you have Skin Diseases at donation site? (Yes/No): no
Do you have Autoimmune Disorders? (Yes/No): no
Do you have Severe Allergic Disorders? (Yes/No): no
Do you have Recipients of Transplants? (Yes/No): n
Please enter either 'Yes' or 'No'
Do you have Recipients of Transplants? (Yes/No): no
Do you have Gonorrhoea, Leprosy, Leishmaniasis, Conjunctivitis (recent)? (Yes/No): no

ADDITIONAL INFORMATION FOR FEMALE DONORS
=====
Do you have Skin Diseases at donation site? (Yes/No): no
Do you have Autoimmune Disorders? (Yes/No): no
Do you have Severe Allergic Disorders? (Yes/No): no
Do you have Recipients of Transplants? (Yes/No): n
Please enter either 'Yes' or 'No'
Do you have Recipients of Transplants? (Yes/No): no
Do you have Gonorrhoea, Leprosy, Leishmaniasis, Conjunctivitis (recent)? (Yes/No): no

ADDITIONAL INFORMATION FOR FEMALE DONORS
=====
Are you currently pregnant? (Yes/No): no
Have you had an abortion in the last 6 months? (Yes/No): no
Are you currently menstruating? (Yes/No): no

=====
ELIGIBILITY CHECK RESULTS
=====
✓ Age check (18-65): PASS
✓ Weight check (45-80 kg): PASS
✓ Hemoglobin check (≥12.5 g/dL): PASS
✓ Blood pressure check (approx 120/80): PASS
✓ Medical history check (no diseases): PASS
✓ Female-specific criteria: PASS

🎉 CONGRATULATIONS! You are ELIGIBLE for blood donation! 🎉

=====
IMPORTANT PRECAUTIONS FOR BLOOD DONATION
=====
1. The donor shall be in good health, mentally alert and physically fit and shall not be inmates of jail or any other confinement.
2. First time donor shall not be over 60 years of age, for repeat donor upper limit is 65 years.
3. The donor shall not be fasting before the blood donation or observing fast during the period of blood donation and last meal should have been taken at least 4 hours prior to donation.
4. Donor shall not have consumed alcohol and show signs of intoxication before the blood donation. The donor shall not be a person having regular heavy alcohol intake.
5. The donor who works as air crew member, long distance vehicle driver, either above sea level or below sea level or in emergency services or where strenuous work is required, shall not donate blood at least 24 hours prior to their next duty shift. The donor shall not be a night shift workers without adequate sleep.

=====
Thank you for your interest in saving lives!
=====

C:\Users\palak\important shi-\python\New folder>
```

Technical difficulty:

Challenge 1: Complex Validation Logic

-Problem: Medical data requires sophisticated validation

- Solution: Implemented comprehensive validation service with medical accuracy

- **Example: Blood pressure validation considering both format and medical ranges**

Challenge 2: State Management

- **Problem: Maintaining donor data across multiple screening steps**
- **Solution: Session management with clear data lifecycle**
- **Result: Robust data handling with proper cleanup**

Challenge 3: User Experience in Console**

- **Problem: Limited UI capabilities in terminal environment**
- **Solution: Progressive form filling with immediate feedback**
- **Innovation: Visual indicators (□/X) for eligibility criteria**

References:

1. World Health Organization (WHO) - Blood Donor Selection Guidelines
 2. American Red Cross - Blood Donation Eligibility Criteria
 3. National Blood Transfusion Services - Standard Operating Procedures
 4. FDA Guidelines for Blood and Blood Products
- Python Official Documentation
 - PEP 8 Style Guide for Python Code